

Project Title : Olist E-commerce Business Performance Analysis

Subtitle : Revenue, Customer Behaviour, Delivery Performance & Customer Satisfaction Analysis using SQL and Power BI

1. Project Overview

This project analyses Olist's e-commerce business performance using SQL and Power BI. The analysis focuses on revenue performance, product concentration, customer purchasing behaviour, delivery reliability, and customer satisfaction.

The objective is to transform raw transactional data into business insights that can support decisions related to revenue growth, customer retention, operational performance, and customer experience.

2. Business Problem

Olist operates a large multi-category e-commerce marketplace, generating substantial transaction, customer, product, payment, delivery, and review data.

However, overall sales performance alone does not explain where the business is performing well or where improvement opportunities exist.

This analysis investigates the business from five perspectives: commercial performance, product concentration, customer retention, delivery performance, and customer satisfaction.

3. Business Questions

3.1 Revenue & Commercial Performance

What is driving Olist's revenue and where are the biggest commercial opportunities?

I investigate ..

Total revenue

Total orders

Average order value

Revenue by product category

Revenue contribution

Category performance

3.2 Product & Commercial Concentration

Is Olist's revenue dependent on a small number of individual products?

I investigate ..

Product-level revenue

Top 10 products

Top 10 revenue contribution

Product concentration

3.3 Customer & Repeat-Purchase Behaviour

Are Olist customers mostly one-time buyers, or is there evidence of repeat purchasing?

I investigate ..

One-time customers

Repeat customers

Revenue contribution

Revenue per customer

Customer retention opportunity

3.4 Delivery & Logistics Performance

How effectively does Olist deliver orders, and where are the biggest delivery-performance issues?

I investigate ..

On-time delivery

Late delivery

Average delivery time

Delivery delay

Delay severity

Monthly delivery performance

3.5 Customer Satisfaction

How satisfied are Olist customers, and are there periods of unusual deterioration in review performance?

I investigate ..

Review score

Positive reviews

Negative reviews

Monthly review trend

Unusual periods of decline

4. Dataset

The project uses the Olist Brazilian E-Commerce Public Dataset, containing information about orders, customers, sellers, products, payments, reviews, and order items.

Dataset/Table

Purpose

Orders	Order dates and delivery information
Order Items	Product sales and revenue
Products	Product categories
Customers	Customer purchasing behaviour
Reviews	Customer satisfaction
Sellers	Seller information
Payments	Payment information

5. Tools & Technologies

Microsoft SQL Server / SSMS

Data validation

Data cleaning

Joins

Aggregation

Customer segmentation

Delivery analysis

Business metric calculations

Power BI

Data modelling

DAX measures

KPI development

Interactive dashboards

Trend analysis

Business visualization

6. Data Preparation

Before analysis, the datasets were validated and prepared using SQL. Data-quality checks included identifying missing values, duplicate records, inconsistent timestamps, and relationships between transactional tables.

Product category values with missing information were handled as unknown to prevent records from being excluded from category-level analysis.

Delivery analysis excluded records where orders were marked as delivered but did not contain the required delivery timestamps.

7. Analysis Approach

Raw Olist Data

--> Data Validation

--> Data Cleaning & Transformation

--> SQL Analysis

--> Business Metrics

-->Power BI Data Model

--> DAX Measures

-->Dashboard

--> Business Insights

--> Recommendations

8. Dashboard

The six dashboard pages are:

1. Revenue & Commercial Performance
2. Product & Commercial Concentration
3. Monthly Revenue Trend
4. Customer & Repeat-Purchase Behaviour
5. Delivery & Logistics Performance
6. Customer Satisfaction & Reviews

8.1 - Revenue & Commercial Performance

Business Question - What is driving Olist's revenue, and where are the biggest commercial opportunities?

Dashboard Overview

The dashboard evaluates overall revenue performance using total revenue, total orders, average order value, and product-category contribution. Category-level visualizations are used to compare revenue performance, order volume, revenue share, and average revenue per order.



Key Findings

- Total revenue was approximately 13.59M across 99,441 orders.
- Average order value was approximately 136.68.
- beleza_saude was the highest-revenue product category.
- Revenue was distributed across a broad range of categories rather than being generated by a single category.
- Some categories generated relatively high revenue despite having lower order volumes, indicating differences in average order value across categories.

Business Interpretation

The results indicate that Olist's commercial performance was driven by a combination of order volume and order value. High-revenue categories should therefore not be evaluated using sales volume alone. Categories with fewer orders can still make a significant contribution when their average order value is higher.

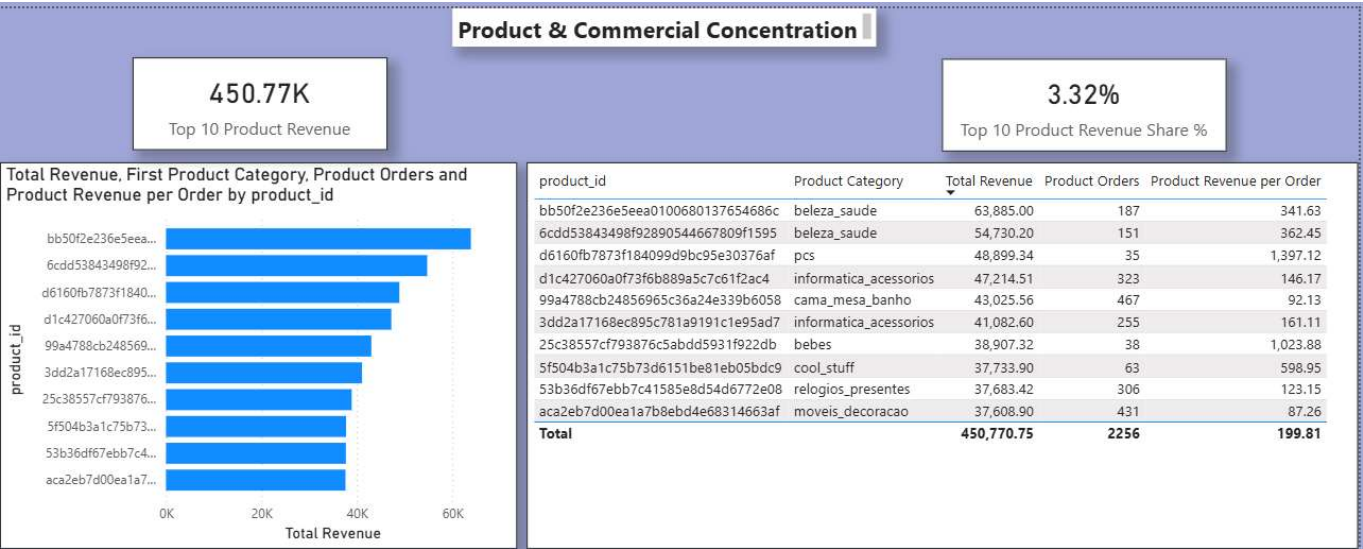
8.2 Product & Commercial Concentration

Business Question

How concentrated is Olist's revenue among individual products, and does the business depend heavily on a small number of products?

Dashboard Overview

The dashboard examines product-level revenue concentration by identifying the highest-revenue products and measuring the contribution of the top 10 products to overall revenue. This helps assess whether Olist's commercial performance is broadly distributed across its product portfolio or concentrated among a relatively small number of products.



Key Findings

The analysis identifies the highest-revenue products and compares their contribution with total product revenue. The Top 10 product concentration metric provides an indication of how much revenue is generated by a relatively small group of products.

Business Interpretation

Product-level concentration provides an additional perspective beyond category-level performance. If a small number of products account for a large share of revenue, these products may require closer attention because availability, pricing, or demand changes could have a disproportionate effect on overall revenue.

Conversely, a lower concentration level would indicate that revenue is distributed across a broader product portfolio, reducing dependence on individual products.

8.3 Monthly Revenue Trend

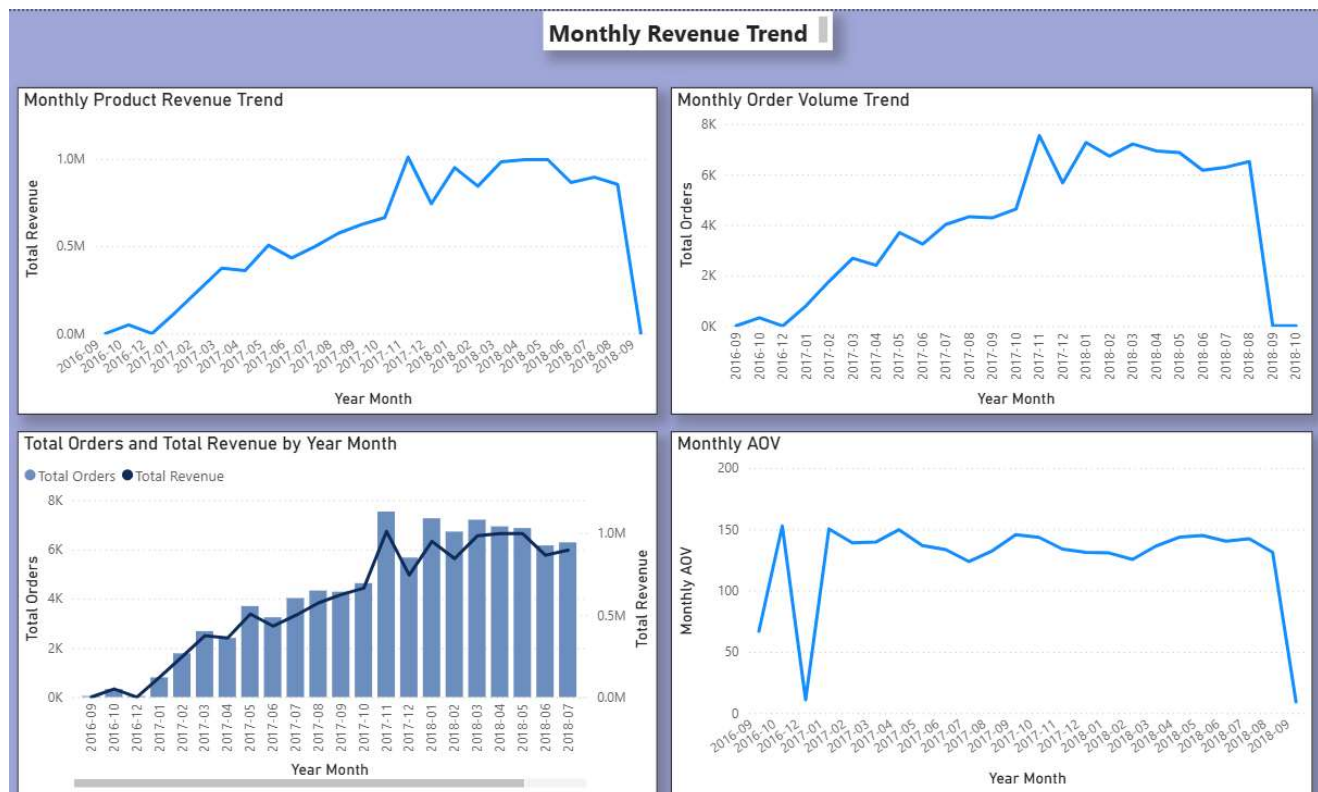
Business Question

How did Olist's revenue and order volume change over time?

Dashboard Overview

This dashboard analyses monthly revenue and order volume to identify changes in commercial activity over time. Comparing revenue with order volume helps determine whether changes were primarily driven by transaction volume or changes in order value.

December 2016 showed an unusually low monthly AOV of 10.90, which was treated as an anomaly requiring further investigation rather than assumed to have the same cause as the review-score decline.



Key Findings

- Revenue generally moved in line with order volume.
- Some months showed differences between revenue growth and order growth.

- The monthly trend highlights periods requiring further investigation.

Business Interpretation

Monitoring monthly revenue together with order volume helps distinguish whether changes in revenue are primarily driven by changes in purchasing volume or by changes in order value.

8.4 Customer & Repeat-Purchase Behaviour

Business Question

Are Olist customers mostly one-time buyers, or is there evidence of repeat purchasing?

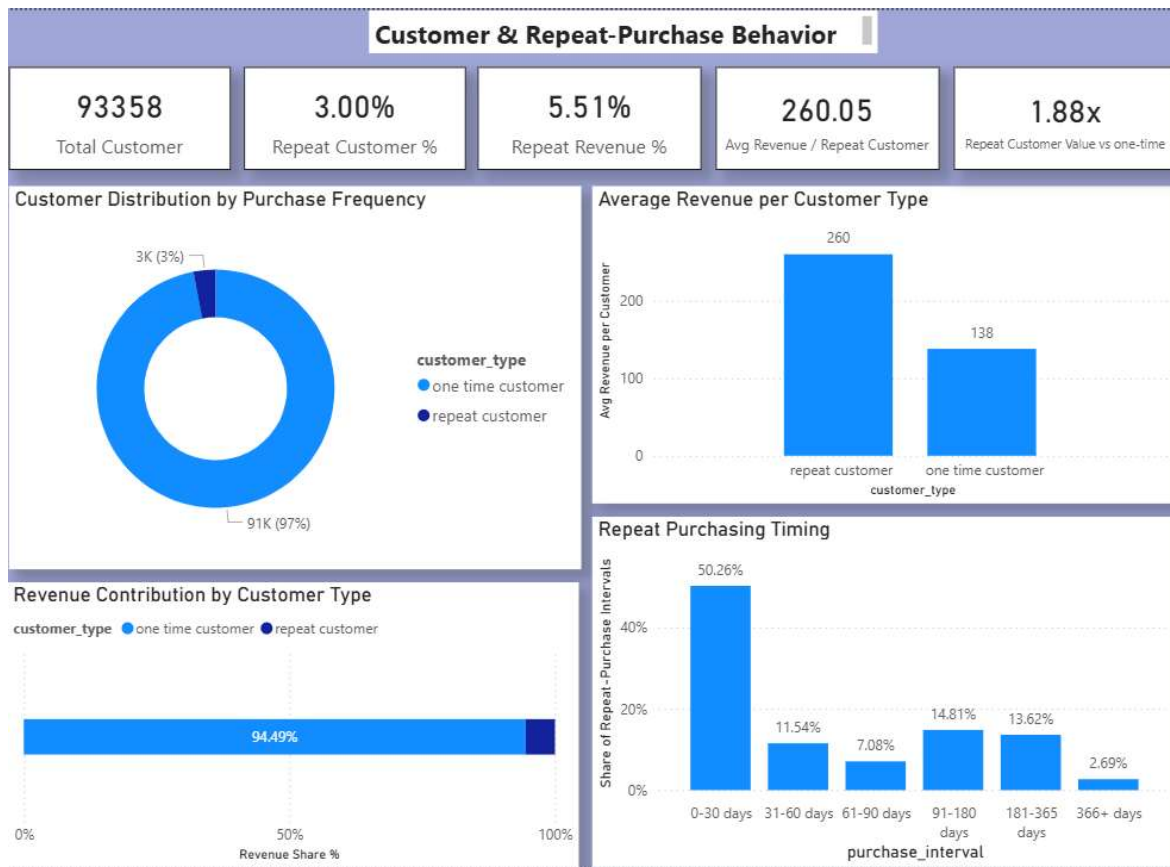
Dashboard Overview

Customer & Repeat-Purchase Behaviour

This dashboard analyses customer purchasing behaviour to understand the balance between **one-time and repeat customers** and their contribution to revenue.

The analysis focuses on:

- One-time vs. repeat customers
- Customer and revenue contribution
- Revenue per customer
- Repeat-purchase behaviour
- Customer retention opportunities



Key Findings

97% of customers were one-time buyers, while only 3% were repeat customers. However, repeat customers generated approximately \$260 per customer compared with \$138 for one-time customers—around 1.9× higher.

Business Interpretation

The low repeat-purchase rate represents a potential retention opportunity. Increasing repeat purchases could improve customer lifetime value and generate additional revenue from existing customers.

8.5 Delivery & Logistics Performance

Business Question - How reliable was Olist's delivery performance, and how frequently were orders delivered late?

Dashboard Overview

This dashboard analyses Olist's delivery performance and logistics reliability, focusing on delivery time, on-time performance, delay severity, and changes in delivery reliability over time.

The analysis includes:

- Delivered orders
- Average and median delivery days
- On-time vs. late orders
- Delay severity
- Monthly on-time delivery trends

The objective is to identify the overall reliability of Olist's delivery operation and highlight areas requiring further investigation.



Key Findings

- 96,470 delivered orders were included in the delivery analysis.
- Average delivery time was approximately 12.5 days.
- Median delivery time was 10 days.
- 93.23% of delivered orders were classified as on time.
- 6.77% were delivered late.
- Among late orders, the largest delay groups were 1–3 days (28.62%) and 4–7 days (27.58%).
- On-time delivery performance declined from 98.88% in 2016 to 92.27% in 2018.

Business Interpretation

Although the overall on-time delivery rate remained above 90%, delivery reliability declined over the analysed period. The increase in delivery issues suggests that logistics performance became an increasingly important operational consideration as order activity expanded.

8.6 Customer Satisfaction & Reviews

Business Question - How satisfied were customers with their purchases, and what relationship exists between customer reviews and operational performance?

Dashboard Overview

This dashboard analyses customer satisfaction through review scores, score distribution, monthly trends, and the relationship between delivery performance and customer reviews.

The analysis focuses on identifying overall satisfaction levels, unusual changes in review scores, and potential operational factors associated with customer dissatisfaction.

1. Review Score

I analysed **99,224 customer reviews** using a 1–5 rating scale.

The overall average review score was:

4.09 / 5

This indicates generally positive customer satisfaction, while the review distribution provides additional insight into dissatisfied customers.

2. Review-Score Distribution

Review Score	Reviews	Share
1 star	11,424	11.51%
2 stars	3,151	3.18%
3 stars	8,179	8.24%
4 stars	19,142	19.29%

5 stars 57,328 57.78%

Key finding: 5-star reviews represented 57.78% of all reviews, while 77.07% were positive (4–5 stars) and 14.69% were negative (1–2 stars).

3. Monthly Review Trend

I analysed the monthly average review score to identify changes in customer satisfaction over time.

December 2016 recorded an unusually low average score of 2.36 / 5. However, the month contained only 45 reviews, so this should be treated as an early-warning signal rather than evidence of a company-wide decline.

4. Reviews & Delivery Performance

Delivery Status Average Review Score

On time 4.29 / 5

Late 2.27 / 5

Late deliveries were associated with substantially lower customer ratings, with a difference of approximately 2.02 review-score points.

This indicates a strong association between delivery reliability and customer satisfaction, although it does not establish causation.

Diagnostic Investigation: December 2016

Further investigation showed that 29 of 45 reviews (64.44%) were 1-star ratings.

Customer comments highlighted recurring issues including:

- Delivery delays and non-delivery
- Stock and fulfilment problems
- Poor communication
- Refund/payment concerns
- Product-quality issues

This suggests that the December decline reflected broader **fulfilment, inventory, and communication problems**, rather than delivery delays alone.



Business Interpretation

Customer review scores remained relatively stable across most months, with December 2016 representing an unusual decline. The limited number of reviews during that month means the result should be treated as an early-warning signal rather than evidence of a sustained deterioration.

Further investigation identified recurring fulfilment, inventory, communication, and product-quality issues, suggesting that the decline was broader than delivery performance alone.

9. Key Insights

9.1 Revenue & Commercial Performance

Olist generated approximately 13.59M in revenue from 99441 orders, with an average order value of approximately 136.68. The beleza_saude category generated the highest total revenue among the product categories analysed.

The category analysis also showed that revenue performance was influenced by both order volume and revenue per order. Some categories generated strong revenue despite having fewer orders, indicating that higher-value purchases can compensate for lower transaction volume.

Key Insight

Olist's revenue performance was driven by a combination of sales volume and order value, rather than order volume alone. This means that commercial performance should be evaluated using both the number of orders and the value generated per order.

Business Implication

Management could prioritize high-revenue categories while also identifying lower-volume categories with relatively high order values. This would help distinguish categories that primarily contribute through transaction volume from those that contribute through higher-value purchases.

9.2 Product Concentration

The product-level analysis examined revenue concentration among individual products, with particular attention to the contribution of the Top 10 products to overall revenue.

The analysis helps determine whether Olist's revenue was broadly distributed across its product portfolio or concentrated among a relatively small number of products.

Key Insight

The Top 10 products contributed approximately 3.32% of total product revenue, indicating that revenue was relatively diversified across individual products rather than heavily concentrated among a small number of products.

Business Implication

The relatively low concentration reduces dependence on a small group of individual products. However, high-performing products should still be monitored for availability, demand, and pricing, while maintaining a diversified product portfolio to reduce product-level revenue risk.

9.3 Customer & Repeat Purchase

The customer analysis showed that Olist's customer base was heavily dominated by one-time purchasers. Approximately 97.00% of customers placed only one order, while just 3.00% were classified as repeat customers.

Despite representing only a small proportion of customers, repeat customers generated approximately 5.51% of total revenue. Their average revenue per customer was approximately 260.05, compared with 137.96 for one-time customers.

Key Insight

Olist had a predominantly one-time customer base, but repeat customers generated substantially higher revenue per customer. This indicates that repeat purchasing customers were more valuable on an individual customer basis.

Business Implication

The low proportion of repeat customers suggests potential for improving customer retention and encouraging additional purchases. Because repeat customers generated higher revenue per customer, increasing repeat purchasing could potentially improve customer lifetime value and revenue efficiency.

9.4 Delivery & Logistics

The delivery analysis covered 96,470 delivered orders. Average delivery time was approximately 12.5 days, while the median delivery time was 10 days.

Overall, 93.23% of delivered orders were classified as on time, while 6.77% were delivered late. Among late orders, the largest delay groups were 1–3 days (28.62%) and 4–7 days (27.58%).

However, delivery reliability declined over the analysed period. The on-time delivery rate decreased from 98.88% in 2016 to 94.35% in 2017 and 92.27% in 2018.

Key Insight

Although the overall delivery performance remained above 90% on time, the declining on-time delivery rate indicates that delivery reliability weakened over time.

Business Implication

The downward trend suggests that logistics performance should be monitored as order activity grows. Particular attention could be given to the causes of delivery delays and to sellers or operational areas associated with consistently poor delivery performance.

9.5 Customer Satisfaction

The customer review analysis showed that review scores were generally stable across most of the analysed period, with a noticeable decline in December 2016.

The decline appears to have been relatively isolated because review performance returned to more typical levels in subsequent periods rather than continuing to deteriorate.

Key Insight

Customer satisfaction did not show a persistent downward trend across the entire period. However, the decline observed in December 2016 represents a period that warrants further investigation.

Business Implication

The December 2016 decline could indicate a temporary operational or customer-experience issue. Potential areas for further investigation include delivery performance, order fulfilment, seller performance, and customer service during that period. The available analysis identifies the pattern but does not establish the specific cause of the decline.

Diagnostic Investigation — December 2016

December 2016 recorded an unusually low average review score of 2.36/5, based on only 45 reviews. Of these, 29 reviews (64.44%) were 1-star ratings, which explains the sharp decline in the monthly average.

Further investigation of customer comments identified recurring complaints involving non-delivery, stock and fulfilment problems, poor communication, refund concerns, and product-quality issues.

The evidence suggests that the December decline reflected a broader customer-experience and fulfilment issue, rather than delivery delays alone. Because review creation date and order purchase date represent different stages of the customer journey, the analysis does not establish a single direct root cause.

The December decline should therefore be treated as an early-warning signal and a case for further operational investigation.

10. Business Recommendations

Based on the findings from the Olist e-commerce analysis, several business opportunities can be identified across commercial performance, product management, customer retention, logistics, and customer satisfaction.

The recommendations below are directly linked to the key findings identified in the analysis and focus on areas where data could support better business decision-making.

10.1 Strengthen High-Performing Product Categories

Finding

Olist generated approximately 13.59M in revenue from 99441 orders, with `beleza_saude` identified as the highest-revenue product category. The analysis also showed that revenue performance varies between categories depending on both order volume and revenue per order.

Recommendation

Prioritize high-performing categories while also identifying categories with strong revenue per order but lower transaction volume. Olist could use category-level performance data to support inventory planning, promotional campaigns, and seller prioritization.

Expected Business Benefit

A more targeted approach to category management could help Olist focus commercial resources on categories with the strongest revenue potential while identifying opportunities to increase sales in promising lower-volume categories.

10.2 Monitor Product Revenue Concentration

Finding

The Top 10 products contributed approximately 3.32% of total product revenue, indicating relatively low concentration at the individual-product level and a broadly diversified product portfolio.

Recommendation

Regularly monitor the contribution of the highest-revenue products and identify products that represent a significant share of total revenue. High-performing products should receive closer attention regarding availability, demand, and pricing.

At the same time, Olist should continue developing a diversified product portfolio to reduce excessive dependence on a small number of products.

Expected Business Benefit

Monitoring product concentration can help reduce revenue risk while ensuring that important high-performing products receive appropriate commercial attention.

10.3 Increase Repeat Purchasing and Customer Retention

Finding

Approximately 97% of customers were one-time buyers, while only 3% were repeat customers. However, repeat customers generated approximately 260.05 revenue per customer, compared with 137.96 for one-time customers.

Recommendation

Olist should investigate strategies that encourage existing customers to make additional purchases. Potential approaches could include personalized product recommendations, targeted promotions, post-purchase engagement, and campaigns designed around previous purchasing behaviour.

Expected Business Benefit

Increasing the proportion of repeat customers could improve customer lifetime value and generate additional revenue from the existing customer base without relying entirely on acquiring new customers.

10.4 Improve Delivery Reliability

Finding

Overall on-time delivery performance was 93.23%, but the on-time rate declined from 98.88% in 2016 to 92.27% in 2018. Late deliveries represented 6.77% of delivered orders.

Recommendation

Olist should monitor delivery performance continuously and identify sellers, regions, and operational processes associated with recurring delays. Delay severity should also be monitored because longer delays may have a greater impact on customer satisfaction.

Seller-level delivery performance can be used to identify consistently strong performers as well as sellers requiring operational attention.

Expected Business Benefit

Improving delivery reliability could reduce customer dissatisfaction, improve the overall customer experience, and help prevent operational issues from becoming more significant as order volume grows.

10.5 Investigate Customer Satisfaction Drivers

Finding

Customer review scores were generally stable across most of the analysed period, but a noticeable decline occurred in December 2016. The analysis identified the pattern but did not establish a specific cause.

Recommendation

Olist should investigate periods of unusual changes in customer satisfaction by comparing review scores with delivery performance, order fulfilment, seller performance, and other operational factors.

The December 2016 decline could be used as a case study for identifying potential relationships between operational performance and customer satisfaction.

Expected Business Benefit

Understanding the factors associated with lower customer satisfaction could help Olist identify operational problems earlier and prioritize improvements that have the greatest potential impact on customer experience.

11. Conclusion

This project analysed the Olist e-commerce dataset to understand the company's commercial performance, product portfolio, customer behaviour, delivery operations, and customer satisfaction. The analysis combined SQL-based data preparation and analysis with Power BI to transform transactional data into business-focused insights and interactive dashboards.

The commercial analysis showed that Olist generated approximately 13.59M in revenue from 99441 orders, with an average order value of approximately 136.68. Revenue performance varied across product categories, with `beleza_saude` identified as the highest-revenue category. The product analysis also provided insight into revenue concentration and the contribution of high-performing products.

Customer analysis revealed a predominantly one-time customer base, with approximately 97% of customers placing only one order. Although repeat customers represented only 3% of customers, they generated higher average revenue per customer than one-time customers. This highlights customer retention and repeat purchasing as an important potential growth opportunity.

Operational analysis showed that 93.23% of delivered orders were on time, while 6.77% were late. However, the on-time delivery rate declined from 98.88% in 2016 to 92.27% in 2018, indicating that

delivery reliability became an increasingly important area to monitor. Customer satisfaction remained relatively stable across most periods, although a noticeable decline in review scores occurred in December 2016 and warrants further investigation.

Overall, the project demonstrates how data analysis can support business decision-making by connecting revenue performance, product concentration, customer behaviour, logistics performance, and customer satisfaction. The analysis identified both strengths and potential improvement areas and translated these findings into practical business recommendations. It also demonstrates my ability to work with SQL for data analysis and Power BI for data visualization, dashboard development, and communicating insights to business stakeholders.

12. Skills Demonstrated

This project demonstrates my ability to apply data analysis techniques to a real-world e-commerce dataset, from data preparation and validation through SQL analysis, visualization, dashboard development, and business recommendations.

12.1 SQL & Data Analysis

- Used SQL Server to explore, validate, clean, and analyse the Olist e-commerce dataset.
- Applied JOIN, GROUP BY, WHERE, CASE, aggregate functions, and Common Table Expressions (CTEs) to answer business questions.
- Used COUNT(DISTINCT ...) to avoid double-counting orders when working with one-to-many relationships.
- Calculated business metrics including revenue, order volume, average order value, customer repeat-purchase behaviour, delivery performance, delay rates, and review scores.
- Performed data-quality checks and investigated missing values, duplicate records, and timestamp inconsistencies.
- Built seller-, customer-, product-, and category-level analyses to support business decision-making.

12.2 Power BI & Data Visualization

- Built interactive Power BI dashboards covering commercial performance, product concentration, customer behaviour, delivery performance, and customer satisfaction.
- Created KPI cards, bar charts, scatter charts, line charts, tables, and trend visualizations.
- Developed measures using **DAX** to calculate business metrics and revenue shares.
- Created and used a date table to support monthly and yearly trend analysis.

- Applied filtering, sorting, relationships, and visual interactions to improve dashboard usability.
- Designed dashboards to communicate analytical findings clearly to business stakeholders.

12.3 Business Analysis

- Translated business questions into measurable analytical metrics.
- Analysed revenue performance by product category and product.
- Evaluated customer purchasing behaviour, including one-time versus repeat customers.
- Investigated delivery reliability, late orders, and delay severity.
- Analysed customer review scores and their changes over time.
- Identified business opportunities related to customer retention, product management, and delivery performance.

12.4 Data Quality & Validation

- Validated analytical results before using them in dashboards and portfolio reporting.
- Identified and handled missing product-category values.
- Investigated duplicate review records and timestamp anomalies.
- Applied consistent rules for delivery analysis, including excluding delivered orders without valid delivery timestamps.
- Cross-checked SQL results against Power BI calculations to improve consistency and reliability.

12.5 Problem Solving

- Investigated and resolved data-model and Power BI relationship issues that affected dashboard calculations.
- Troubleshoot incorrect visual results caused by inactive relationships and aggregation behaviour.
- Considered one-to-many relationships when designing seller and customer analyses to avoid inflated metrics.
- Used a step-by-step validation approach to confirm individual metrics before combining them into broader analyses.

12.6 Data Storytelling & Communication

- Structured the analysis around clear business questions rather than presenting metrics without context.
- Converted analytical results into key insights and business recommendations.
- Designed dashboard pages to communicate commercial, customer, operational, and satisfaction-related findings.
- Presented findings using the structure Business Question > Analysis > Key Finding > Business Implication.

12.7 Tools

SQL Server / SSMS — Data querying, validation, transformation, and analysis

Power BI — Data modelling, DAX, visualization, dashboard development, and reporting

Microsoft Word — Portfolio documentation and presentation